

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

End Semester Winter Examination – Nov. 2019

Course: B. Tech in -Civil Engineering

Sem: IV

Subject Name: Hydraulic Engg. II

Subject Code: BTCVC401

Max Marks: 60

Date:- 26/11/19

Duration:- 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Attempt **any five** questions of the following.
3. Illustrate your answers with neat sketches, diagram etc., wherever necessary.
4. If some part or parameter is noticed to be missing, you may appropriately assume it and should mention it clearly

Marks

Q. 1 Attempt any two of following Questions

- (A) Explain classification of flow in open channel **and** Types of channels. 6
- (B) Water flows over a rectangular weir 1.5m wide at a depth of 160mm and afterwards passes through a triangular right-angled weir. Taking Cd for the rectangular and triangular weir as 0.62 and 0.59 respectively. Find the depth over the triangular weir. 6
- (C) Define most economical section of channel. Derive condition for most economical rectangular channel. 6

Q.2 Attempt following Questions

- (A) Calculate the quantity of water that will be discharged at a uniform depth of 0.9m in a 1.2m diameter pipe which is laid at a slope 1 in 1000. Take $C=55$ 4
- (B) Explain Specific energy and Specific energy curve. 4
- (C) The specific energy for a 4 m wide rectangular channel is to be 3 Nm/m. If the rate of flow of water through the channel is $18 \text{ m}^3/\text{s}$. Determine the alternate depths of flow. 4

Q. 3 Attempt following Questions

- (A) Derive the expression of depth of hydraulic jump. 6
- (B) Give classification of channels according to flow (only table) and derive expression for the length of back water curve / Direct step method. 6

Q.4 Attempt following Questions

- (A) Explain velocity triangles at outlet and inlet of unsymmetrical moving curved 8

plate when jet strikes tangentially at one of the tips. Derive the expression for

- 1) The force exerted by jet of water on same plate
- 2) Work done per second per unit weight of fluid striking per second
- 3) Work done per second per unit mass of fluid striking per second.
- 4) Efficiency of jet

(B) A nozzle of 50mm dia. delivers a stream of water at 20 m/s perpendicular to a plate that moves away from the jet at 5 m/s. Find **4**

- 1) The force on the plate
- 2) The efficiency of the jet.

Q.5 Attempt following Questions

(A) Elaborate major component parts of pelton wheel, Francis and Kaplan Turbine with neat sketch. **4**

(B) A pelton wheel has a mean bucket speed of 10 m/s with a jet of water flowing at the rate of 700 lit/sec under a head of 30m. The bucket deflects the jet through an angle of 160° , Calculate the power given by water to the runner and the hydraulic efficiency of the turbine. Assume coefficient of velocity as 0.98. **4**

(C) Explain **4**

- 1) Efficiencies of a Centrifugal pump
- 2) Airlift pump and submersible pump.

Q.6 Attempt following Questions

(A) Explain – Laminar boundary layer, Turbulent boundary layer, Laminar sub layer, Boundary layer thickness, Displacement thickness, Momentum thickness, Energy thickness and coefficient of Drag. **8**

(B) Determine the thickness of a boundary layer at the trailing edge of smooth plate of length 4m and of width 1.5m, when the plate is moving with a velocity of 4 m/s in stationary air. Take kinematic viscosity of air as $1.5 \times 10^{-5} \text{ m}^2/\text{s}$. **4**

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